

AIML Capstone 1: Autonomous Driving

Part 1: vehicle object detection (Faster R-CNN). Part 2: Tesla Autopilot and road-safety EDA.

1. Business context

AV systems need real-time detection of vehicles, pedestrians, and cyclists. Part 1 builds a bounding-box detector; Part 2 analyses fatal Tesla incidents.

2. Part 1 — Object detection

Dataset. 5,626 images + labels.csv (11 classes, xyxy boxes).

Model. Faster R-CNN ResNet-50 FPN (COCO pretrained), custom classification head.

Training. 90/10 split, 1,500-image subset, SGD, 3 epochs, batch 4.

Evaluation. Validation mean IoU and qualitative inference plots.

3. Part 2 — Tesla deaths EDA

Dataset. 307 rows cleaned to 303 accident records.

- 247 accidents with 1 death; Tesla driver fatalities sum: 2,371.
- 47.5% of events had Tesla driver/occupant death.
- 49 cyclist/pedestrian events; 38% involved other-vehicle fatalities.
- Top models: Unknown (188), S (45), 3 (35), X (22), Y (13).

4. Tools

Part 1: PyTorch/torchvision (GPU/Colab). Part 2: pandas, matplotlib, seaborn.

5. Conclusion

Detection pipeline plus safety analytics address the autonomous-driving scenario.